AI-Based Pet Behavior Analysis System Using NLP and Machine Learning

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FYP Proposal

Introduction

In today's growing pet care landscape, recognizing the early signs of illness in companion animals is essential to ensuring their wellbeing. Behavioral changes in pets often serve as early indicators of underlying health issues or diseases. Veterinary research has shown that unusual behaviors such as aggression, loss of coordination, excessive drooling, or changes in activity levels can signal medical conditions that require early intervention (Avocadriveanimalhospital.com.au, 2024). However, pet owners often don't have access to the proper tools to efficiently track and interpret these behavioral patterns over time, which can delay the identification of potential health concerns. (Pubudu Senanayake, Samanthi Wickramsinghe and Sunesh Hettiarachchi, 2021)

This is especially significant for serious, fast-progressing conditions such as rabies, parvovirus, and distemper, where early behavioral and physical signs may appear well before a case becomes severe. Without a consistent way to record and make sense of these observations, subtle warning signs can go unnoticed, leading to delayed treatment.

To address this challenge, this project proposes Pawlytics, a pet behavior-analysis system that helps owners interpret what they observe in their pets. An owner describes their pet's behavior or symptoms in everyday language, and Pawlytics uses Natural Language Processing (NLP) and Machine Learning to classify the description into a health-related category, returning a possible cause, a confidence score, and a recommended action. By turning unstructured behavioral observations into structured, data-driven insight, Pawlytics aims to support earlier detection of potential health issues.

Problem Statement

Pawlytics will be a very useful application for both pet owners and vets in Sri Lanka and these are a few reasons why.

1.1. Pet owners

Preventable Diseases not being detected early

Many diseases in pets can be prevented or treated effectively if they are identified at an early stage. However, pet owners often fail to recognize the early symptoms or behavioral changes that indicate underlying health issues. According to veterinary research, early signs of diseases such as infections, neurological conditions, or rabies may appear through subtle behavioral or physical symptoms before they become severe. Without a system to consistently track symptoms, vaccination schedules, and health records, these early warning signs may go unnoticed, leading to delayed diagnosis and treatment. (epid.gov, 2025)

Limited access to Behaviour-based Health Insights

Behavior is a key indicator of an animal's health and wellbeing. Changes in behaviour such as aggression, unusual lethargy, loss of coordination, excessive drooling, or appetite changes may indicate underlying medical or neurological conditions. However, most pet owners do not have access to tools that help them record, interpret, and understand behavioral patterns over time. (The and The, 2025)

1.2. For vets and clinics

No Easy Access to historical records

When a pet is brought in, the vet depends heavily on the owner's account of what happened before the visit. Owners frequently struggle to recall exactly what they observed, how often, and when it started, and important behavioural details between visits are lost or reported inconsistently. This incomplete and unstructured history makes it harder to understand how symptoms developed and can slow accurate assessment.

Proposed Solution

This is the proposed solution to overcome the issues mentioned in the above problem statement:

1. **AI-Driven Behavior Analytics Module**

- Pet owners can log behavioral changes or symptoms observed in their pets.
- The system uses NLP to interpret symptom descriptions entered by owners.
- Machine learning models analyze behavior patterns using historical health and behavior data and will provide a response to pet owners.
- Generate insights and recommendations on potential health concerns.
- For each input the system returns a possible cause, a confidence score, and a recommended action with an urgency level (for example: monitor, consult a vet soon, or seek urgent care).

2. **Behaviour Logging and Pattern Tracking**

- Every logged observation and its prediction are stored, building a behaviour history for each pet.
- The system detects repeated or escalating patterns for example, the same concerning behaviour recurring over several days.

3. **Analytical Dashboard**

- An interactive dashboard presents the behaviour log, a timeline of observations per pet, category and urgency breakdowns, and alerts for repeated patterns.
- The structured, time-stamped history can be reviewed at a consultation.

Feasibility

Technical Feasibility

The backend will be developed using Python (Flask/FastAPI), which supports API development and easy integration with machine learning models. For NLP and behavior analysis, libraries such as HuggingFace, spaCy, and scikit-learn can be used to process and analyze text-based pet records.

A PostgreSQL database will store structured data such as pet profiles, vaccination history, and behavioral logs, while React.js will be used to build a simple and interactive web interface for users.

From an operational point of view, the system will mainly support the pet owners as primary users, with veterinarians having access to health records for reference and history tracking.

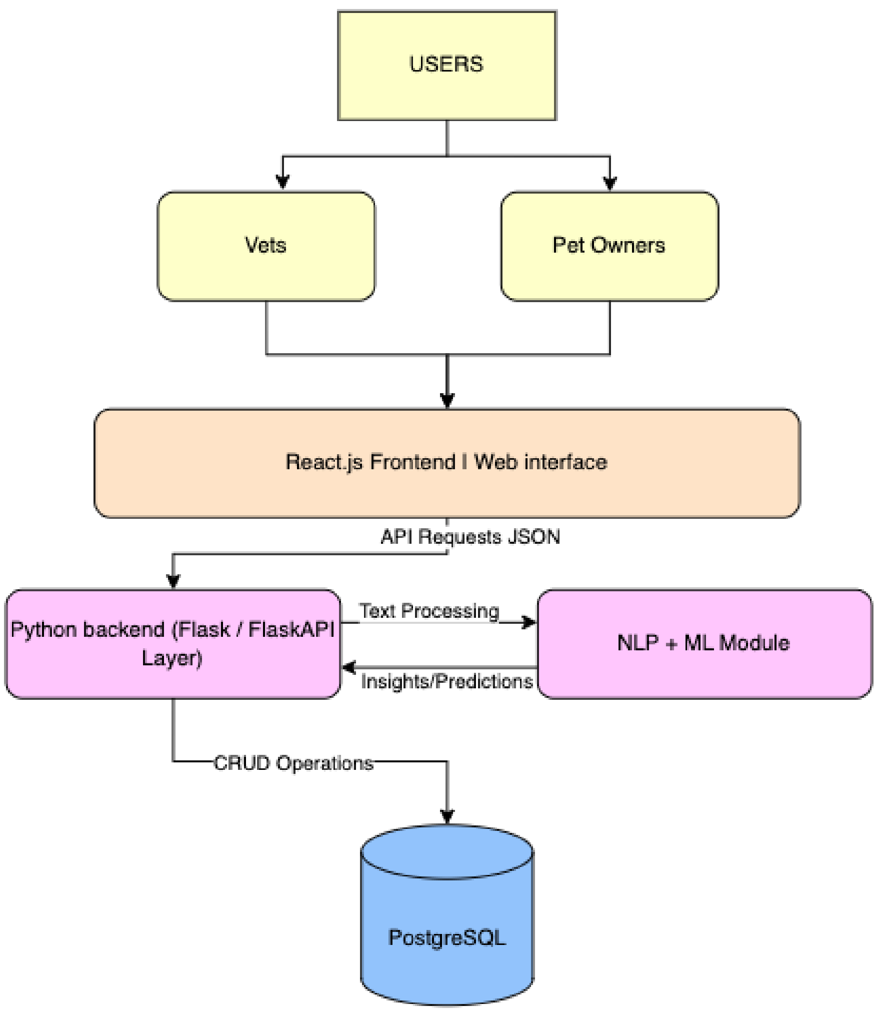
Data Feasibility

The data required for testing and training the model can be obtained via academic papers and veterinary knowledge bases.

Project Description

The main goal of this system is to improve pet healthcare monitoring by helping pet owners detect potential issues early on and keep track of all records through a simple digital platform.

System Architecture



Deliverables

The models will be built and trained in Google Colab, which runs in the browser and gives free GPU access. This means no machine-learning packages need to be installed locally, which solves the problem of working on an older laptop. The work uses scikit-learn for the classical models and metrics, HuggingFace and sentence-transformers for the embedding and transformer models, and spaCy or NLTK for text preprocessing.

The web interface and dashboard will be built with Streamlit, which lets the whole app be written in Python without a separate front end or a full-stack setup. Logged observations and predictions are kept in a lightweight store, which is enough here and avoids the overhead of a full database.

Additionally, the Agile Sprint Methodology will be followed during development to achieve a faster and more efficient outcome.

In Scope

These are the high level overviews of the features which will be included in the scope for this project delivery or the Minimum Viable Product (MVP).

User Management Features

Role Based Accesses for:

- Admins
- Pet Owners

Pet Management

- Pet Registration
- Pet Profile Management

Behavior & Symptom Log

- Pet owners can log daily behavior changes or symptoms
- Store historical behavior data linked to each pet
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Behavior Analytics Module (AI-Based)

- Analyze logged symptoms and behavior descriptions using NLP.
- Classify behavior into categories.

- Generate basic health insights and recommendations.
- Provide alerts for unusual or repeated behavioral patterns.

Dashboard

- Behaviour log and timeline per pet
- Category and urgency breakdowns, with pattern alerts
- A model-comparison view showing performance across the trained models

Software Artifacts

- Fully functional backend API with authentication and data validation
- Working frontend web application with user interface for pet owners and vets
- Trained NLP and machine learning models for behavior classification
- Complete source code repository with documentation

Analytical Outputs

- System design documentation and architecture diagrams
- Database schema and data model specifications
- Machine learning model evaluation reports (accuracy, precision, recall metrics)
- Comparative analysis of behavior classification performance

Investigation and Evaluation

- Literature review on pet health informatics and behavior analysis
- Case studies demonstrating system functionality
- Critical evaluation of system limitations and challenges
- Recommendations for production deployment and future enhancements

Resources Required

The project will be developed using the following tools and technologies:

Software / Tools

- Python (Flask / FastAPI) – backend development
- React.js – frontend development
- PostgreSQL (Supabase / Neon) – cloud database
- Jupyter Notebook / Google Colab – model development and testing

Libraries

- spaCy / NLTK – NLP processing
- scikit-learn – machine learning models
- pandas / NumPy – data handling
- Matplotlib – data visualization

Other Tools

- GitHub – version control
- VS Code – development environment

Expectations

Expected Output

- Centralized access to pet medical records, vaccinations, and behavior logs
- An AI-powered behavior analysis module that processes symptoms and behavior descriptions
- A dashboard displaying health history and basic AI-generated insights – Out of scope for MVP.

Expected Outcome:

- Improved pet medical record management by replacing manual vaccination books with a digital system.
- Better continuity of care, as veterinarians can access complete pet histories in one place
- Increased awareness among pet owners about early signs of illness through behavior tracking.
- Faster identification of potential health risks using AI-based analysis of symptoms and behavior patterns
- Reduced dependency on incomplete records or informal online advice for pet health decisions

Timelines

These are the estimated timelines for the completion and full deliverable of this project.

Phase	Duration	Key Deliverables
Phase 1: Design & Setup	1 week	Architecture design, schema, dev environment setup
Phase 2: Backend Development	3.5 weeks	API endpoints
Phase 3: NLP & ML Integration	3.5 weeks	Model training, behavior classification, inference
Phase 4: Frontend Development	2.5 weeks	Wireframes, Web UI, user workflows
Phase 5: Testing & Refinement	1 week	Unit tests, integration tests, bug fixes
Phase 6: Documentation & Evaluation	1 week	Final documentation, analysis, case studies

Limitations

- The system will be relying on user-entered behavioral and symptom data, which may be subjective and vary in accuracy depending on the pet owner's observations and inputs.
- The machine learning model will be trained on a limited dataset within an academic scope, which may affect prediction accuracy and generalization to real-world cases in Sri Lanka.
- The system provides supportive insights only and does not replace professional veterinary diagnosis or medical judgment. It is not to be used a substitute for vets, and pet owners will be escalated to vets where necessary.
- Availability of high-quality labelled veterinary behavioral datasets is limited, which may restrict model performance and depth of analysis.

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